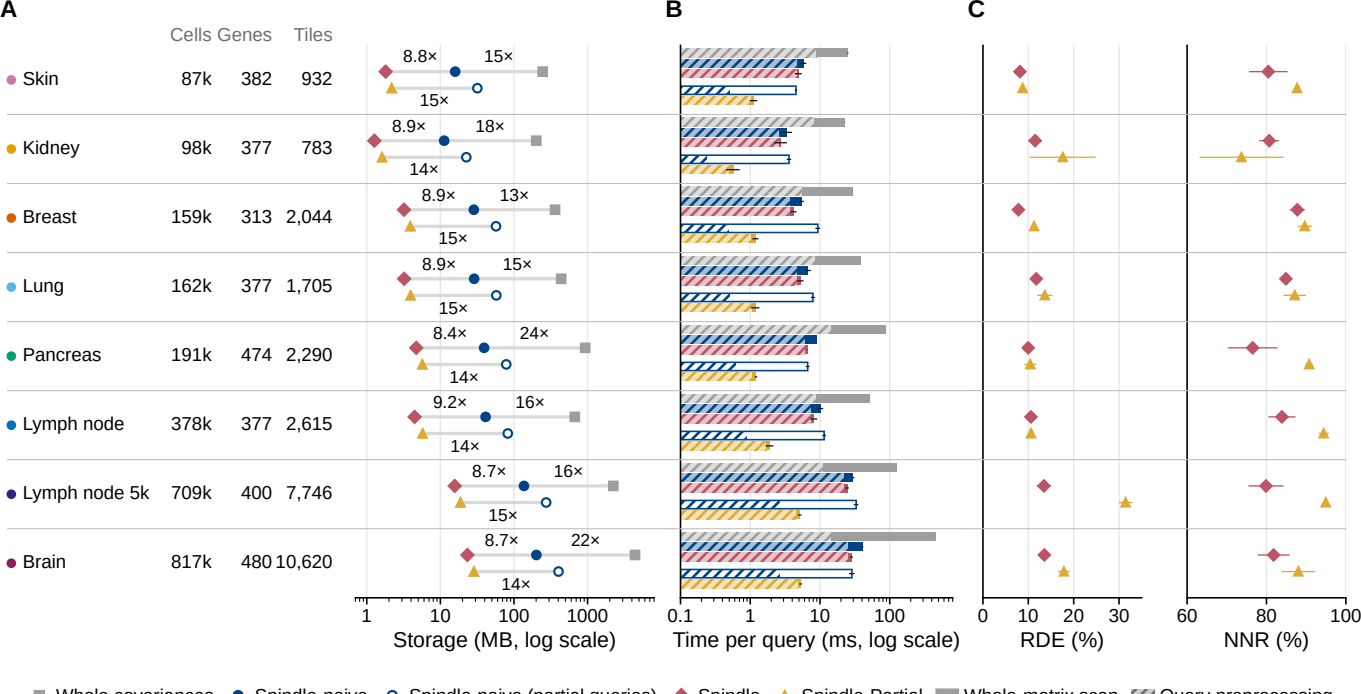
A
B
C
Cells Genes Tiles
Skin 87k 382 932
Kidney 98k 377 783
Breast 159k 313 2,044
Lung 162k 377 1,705
Pancreas 191k 474 2,290
Lymph node 378k 377 2,615
Lymph node 5k 709k 400 7,746
Brain 817k 480 10,620
8.8× 15× 15×
8.9× 18× 14×
8.9× 13× 15×
8.9× 15× 15×
8.4× 24× 14×
9.2× 16× 14×
8.7× 16× 15×
8.7× 22× 14×
1 10 100 1000
Storage (MB, log scale)
0.1 1 10 100
Time per query (ms, log scale)
0 10 20 30
RDE (%)
60 80 100
NNR (%)
Whole covariances
Spindle-naive
Spindle-naive (partial queries)
Spindle
Spindle-Partial
Whole-matrix scan
Query preprocessing